

Project 04: Autonomous SQL ReAct Agent

Tool-Augmented Reasoning & Acting Agent Loop

1. Executive Overview

The ReAct SQL Agent implements the Reasoning + Acting paradigm. The agent dynamically decides which database tools to invoke (inspect_schema, validate_sql, run_sql_query), observes output results, self-corrects invalid queries, and synthesizes final answers.

2. System Architecture & Workflow Pipeline

- Thought 1: Inspect database schema -> Action: inspect_schema()
- Observation 1: Retrieve tables & column data types
- Thought 2: Formulate SQLite SELECT query -> Action: run_sql_query(sql)
- Observation 2: Execute query safely & capture tabular rows
- Synthesis: Generate final answer with full trajectory log

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves ReAct SQL Agent web UI
GET	/api/schema	Returns database schema specification
POST	/api/run-agent	Executes ReAct agent workflow and returns trajectory logs

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 04_sql_agent/src && python -m uvicorn server:app --host 127.0.0.1 --port 8004
```

Automated Test Suite Execution:

```
python -m pytest 04_sql_agent/tests/
```